

VS Code Reference (Optional)

Optional background - not required for HW1

What is VS Code?

Think of VS Code as a well-equipped workshop:

- **Tools** - Built-in coding features
- **Tool Storage** - Extensions for specialized tasks
- **Workbench** - Clean, organized workspace
- **Magnifying Glass** - Debugging & analysis
- **Reference Books** - Integrated documentation

Why Choose VS Code?

- **Free & Open Source** - No licensing fees, community-driven
- **Lightweight but Powerful** - Fast startup, rich features
- **Language Support** - Works with 50+ languages
- **Extensible** - 40,000+ extensions to customize your workflow
- **Built-in Tools** - Git, terminal, and debugger
- **Cross-Platform** - Same development environment on Windows, macOS, and Linux

What Can VS Code Do?

- **Smart Editing** - Syntax highlighting, IntelliSense, snippets
- **Code Navigation** - Go to definition, find references, symbol search
- **Debugging** - Breakpoints, step-through, variable inspection
- **Version Control** - Git integration, visual diffs, commit management
- **Project Management** - Multi-folder workspaces, file explorer, integrated terminal

Command Palette & IntelliSense

- **Command Palette:** `Ctrl+Shift+P` / `Cmd+Shift+P` - one search bar for all VS Code features, fuzzy search
- **IntelliSense**, just like predictive text on your phone:

```
pr -> print()
```

```
console. -> log() | error() | warn()
```

Example: Python Project

1. Open Folder: `my_first_project`

2. Create File: `hello.py`

3. Write Code:

```
print("Hello, VS Code!")  
name = input("Your name: ")  
print(f"Nice to meet you, {name}!")
```

4. Run: Right-click -> *Run Python File in Terminal*

Debugging Basics

- **Breakpoints** - Stop at specific lines
- **Step Through** - F10 (over), F11 (into), F5 (continue)
- **Inspect Variables** - Hover to see values, use Watch panel

You'll need language-specific extensions (like PHP Debug) installed to debug that language.

Customizing VS Code

Themes: File -> Preferences -> Color Theme

Keyboard Shortcuts: File -> Preferences ->

Keyboard Shortcuts

Settings: File -> Preferences -> Settings

(Ctrl+,)

- **Project-specific:** `.vscode/settings.json` in your folder
- **Global:**
 - Mac: `~/Library/Application Support/Code/User`
 - Windows: `%APPDATA%\Code\User`

Git Integration & Search

- **Stage Changes:** Click  next to files
- **Commit:** Add a message, click checkmark
- **Push/Pull:** Sync with GitHub or other repos
- **Find in File:**  / 
- **Find in All Files:**  / 
- **Quick Open File:**  / 

Essential Shortcuts

Action	Windows/Linux	macOS
Command Palette	Ctrl+Shift+P	Cmd+Shift+P
Quick Open	Ctrl+P	Cmd+P
Toggle Terminal	Ctrl+`	Cmd+`
Comment Code	Ctrl+/*	Cmd+/*
Format Document	Shift+Alt+F	Shift+Opt+F

VS Code Extensions

Supercharge Your Editor

Extensions: Add New Powers

Need	Extension
Python Dev	Python, Pylance
Web Dev	Live Server
Git Tools	GitLens
Formatting	Prettier
Themes	Material Theme

How to Install

- **Marketplace:** Extensions icon -> Search -> Install
- **Command Palette:** `Ctrl+Shift+P` -> "Extensions: Install"
- **CLI:**

```
code --install-extension ms-python.python  
code --install-extension esbenp.prettier-vscode
```

Key Extensions by Category

Python

Python · Pylance · Jupyter · Black Formatter

Web Development

Live Server · HTML CSS Support · ESLint · Emmet · Auto
Rename Tag

Git & Version Control

GitLens · Git Graph · GitHub Pull Requests

Debugging & Testing

Jest · Test Explorer UI

Code Quality & Formatting

Prettier · SonarLint

Productivity

GitHub Copilot · Bracket Pair Colorizer · Path Intellisense ·
TODO Highlight

| Remote Development

Remote - SSH · Remote - WSL · Dev Containers

| Database & API

SQLTools · MongoDB for VS Code · PostgreSQL · REST Client

| Documentation

Markdown All in One (Markdown editing helpers - not a Marp slide renderer) · Draw.io Integration · Better Comments

| Team Collaboration

Live Share · CodeTour · Settings Sync

Top 10 Extensions for Software Engineers

1. **Python** - Core Python support
2. **Pylance** - Better IntelliSense
3. **Live Server** - Instant web preview
4. **Prettier** - Code formatter
5. **ESLint** - JavaScript/TypeScript linting
6. **GitLens** - Git history & blame (or Git Graph / Git History)
7. **Bracket Pair Colorizer** - Easy code navigation
8. **Path Intellisense** - File path autocomplete